COLLEGE OF SCIENCE AND TECHNOLOGY

GAKO CAMPUS

COMPUTER ENGINEERING LEVEL III

MODULE: MOBILE APPLICATION

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ASSIGNMENT: LAB 5 Posts Manager with SQLite

This project is an Offline Posts Manager developed using Flutter and SQLite. It allows users to perform CRUD operations on posts without internet connection. The application demonstrates local database integration and asynchronous data handling in Flutter.

HAND WRITTEN SCANNED DOCUMEND

post Manager with Sqlite

1. Dependencies I used and the reasons why

In this project, I used the following dependencies:

> sqflite : this is used to work with SQLite database in Flutter.
it allows us to create, read, update, and delete data.

> path : helps to define the correct path where the database file is stored on the device.

> image_picker : is used to select images from the gallery and attach them to posts.

The reasons Why SQlite is necessary

SQlite is important because it allows us to store data locally on the device.

▷ the app works without internet (offline).
▷ Data is saved permanently (not lost when app closes)
▷ it is fast and lightweight
▷ Good for small app like posts manager

So, SQlite makes our app reliable even when there is no ~~t~~ no network.

2. How I handle database exceptions

in my project, I handle errors using try-catch blocks to prevent the app from ~~to~~ crashing.

▷ Handle database not initialized

→ if the database is not ready, I ensure it is initialized first before using it.

Example idea:

- Call `initDatabase()` before any operation
- ~~the~~ check if database is null

▷ Handle insert/update/delete errors

when inserting, updating, or deleting:

- I use try-catch
- if an error happens, I show message or print the error

Example:

dart

```
try {
  await db.insert(...);
} catch (e) {
  Print("Insert error: $e");
}
```

▷ Handle invalid or corrupted data

- I validate inputs (title and body should not be ~~empt~~ empty)
- I check if data exists before updating or deleting
- I avoid null values that can crash the app

3. Explain SQlite in Flutter

Database Vs Table

> Database: A container that stores all data
  Example: (posts.db)

> Table: A structure inside the database that stores records.
  Example: (posts) table with title, body, date, image)

So:

- Database = whole storage
- Table = data ~~organ~~ organized in rows and columns.

CRUD operations:

CRUD means :

> Create -> Add new post
> Read -> Get all posts
> Update -> Edit a posts
> Delete -> Remove records

Example in our app:

- Add post -> insert into database
- View posts -> fetch from database
- Edit -> update data
- Delete -> remove record

How Flutter interacts with database (Asynchronous)

Flutter uses async/await because database operations take time.

Example :

```
await DatabaseHelper.instance.getPosts();
```

Why async?

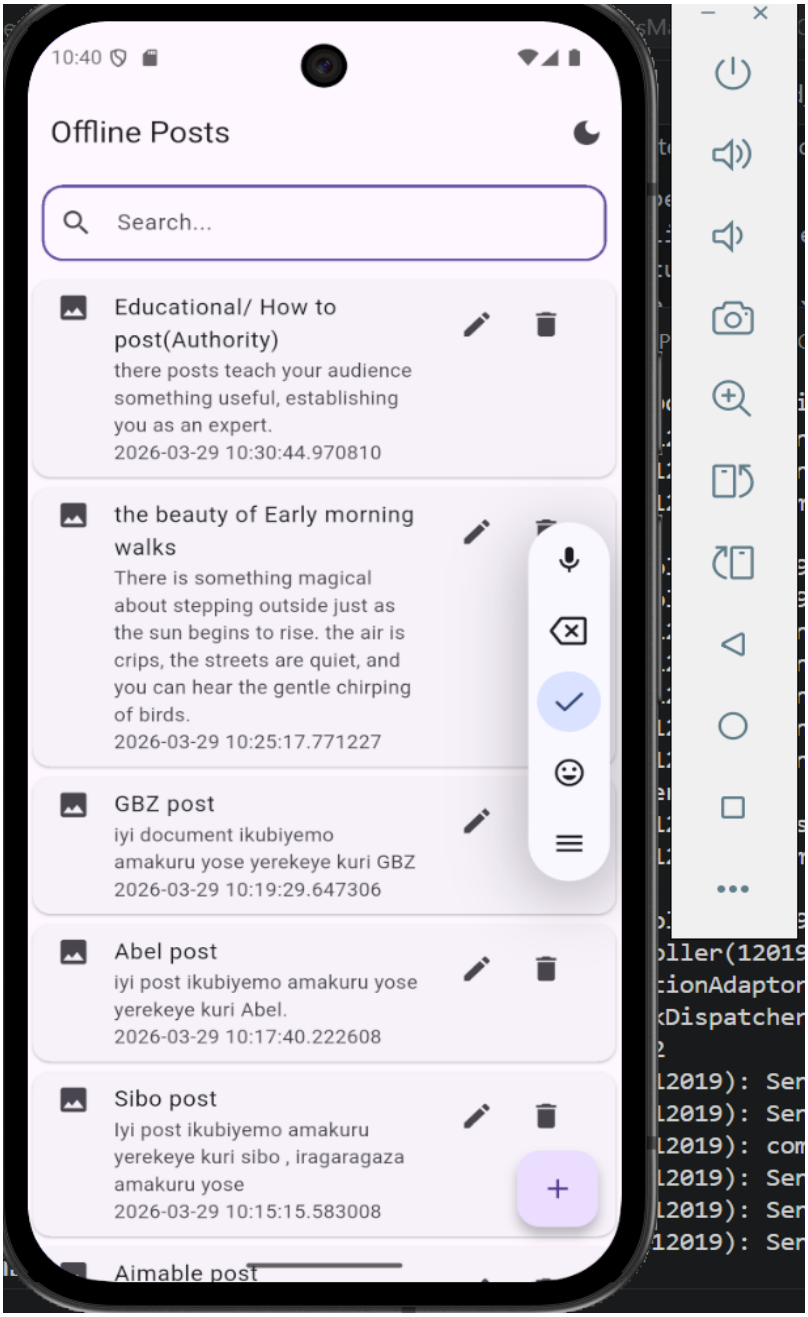
-> it prevents the app from freezing
-> keeps UI smooth
-> allows background operations

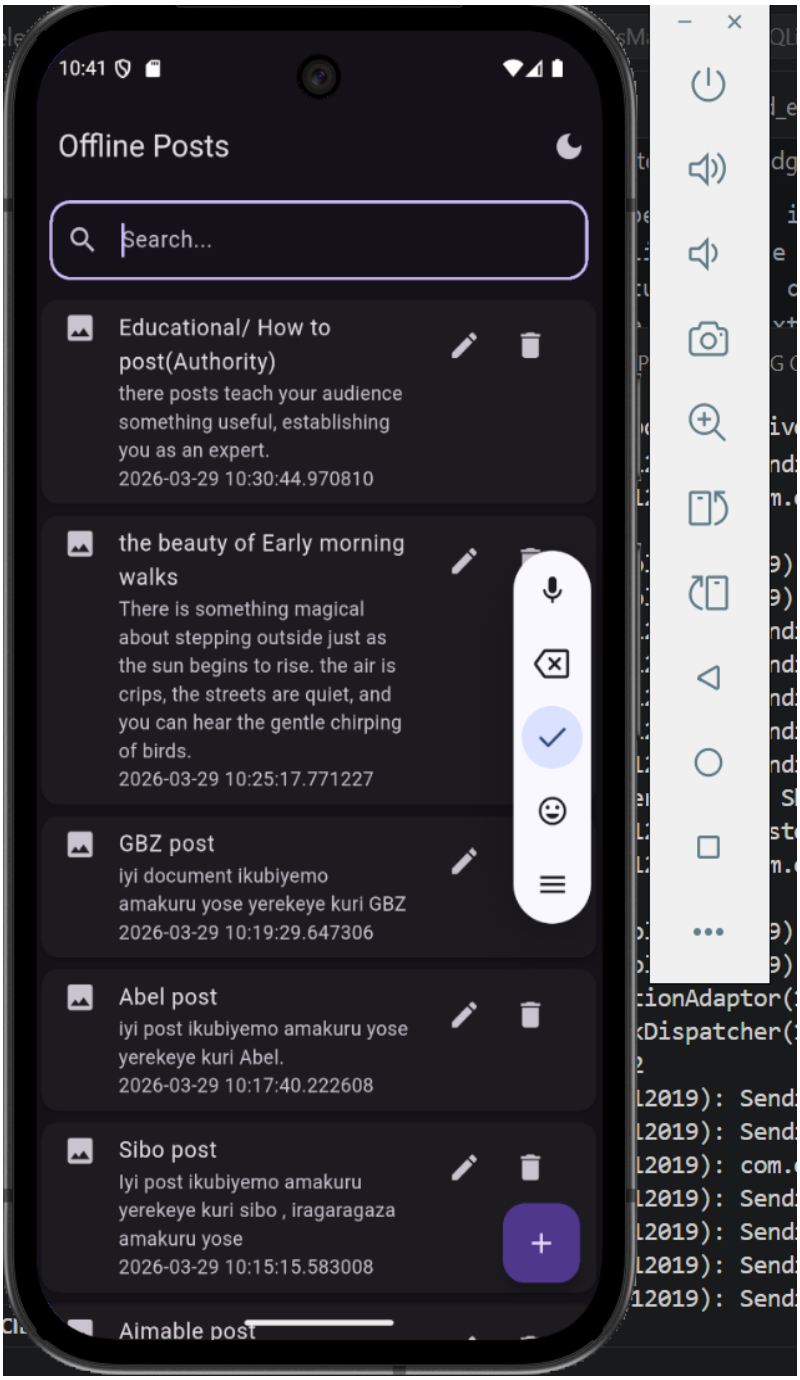
So flutter sends a request, waits for the result, then updates the UI

Posts Manager with SQLite screenshot

1. Home Screen

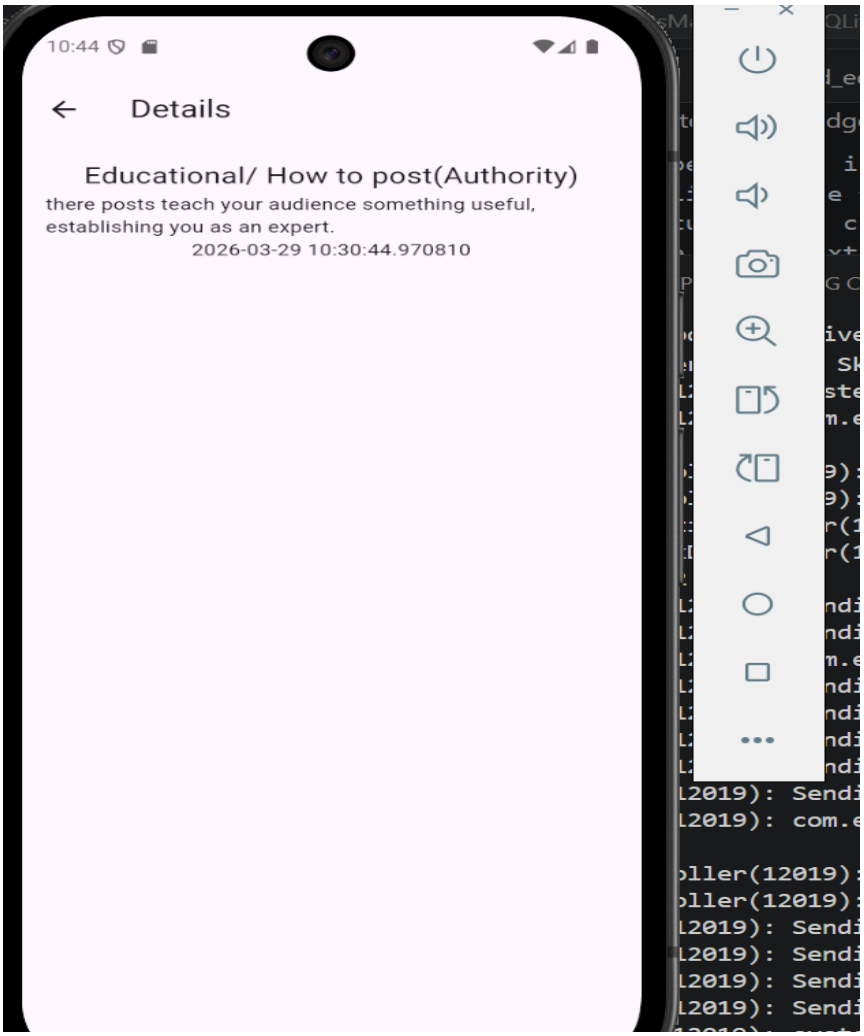
This screenshot shows all posts stored locally

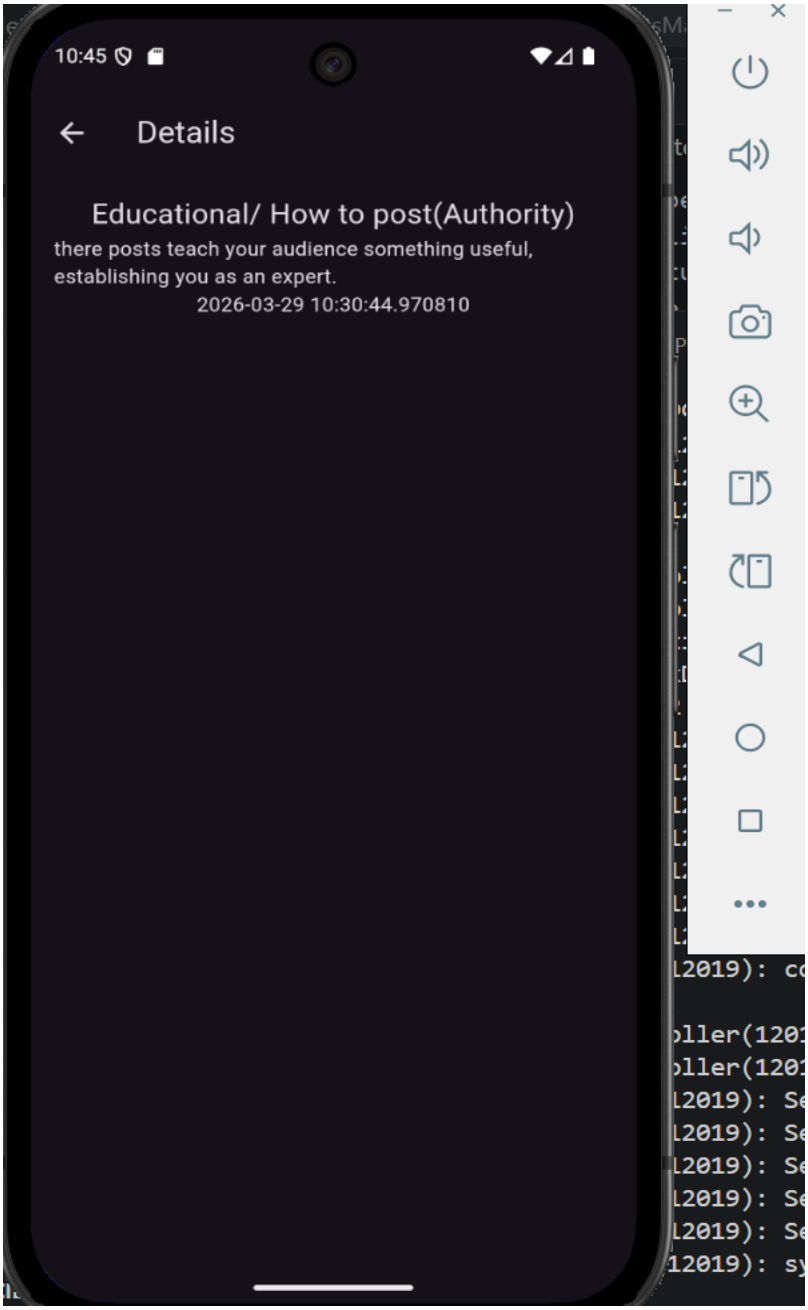



10:41
Offline Posts
Search...
Educational/ How to post(Authority)
there posts teach your audience something useful, establishing you as an expert.
2026-03-29 10:30:44.970810
the beauty of Early morning walks
There is something magical about stepping outside just as the sun begins to rise. the air is crips, the streets are quiet, and you can hear the gentle chirping of birds.
2026-03-29 10:25:17.771227
GBZ post
iyi document ikubiyemo amakuru yose yerekeye kuri GBZ
2026-03-29 10:19:29.647306
Abel post
iyi post ikubiyemo amakuru yose yerekeye kuri Abel.
2026-03-29 10:17:40.222608
Sibo post
Iyi post ikubiyemo amakuru yerekeye kuri sibo , iragaragaza amakuru yose
2026-03-29 10:15:15.583008
Aimable post

2. Post Details

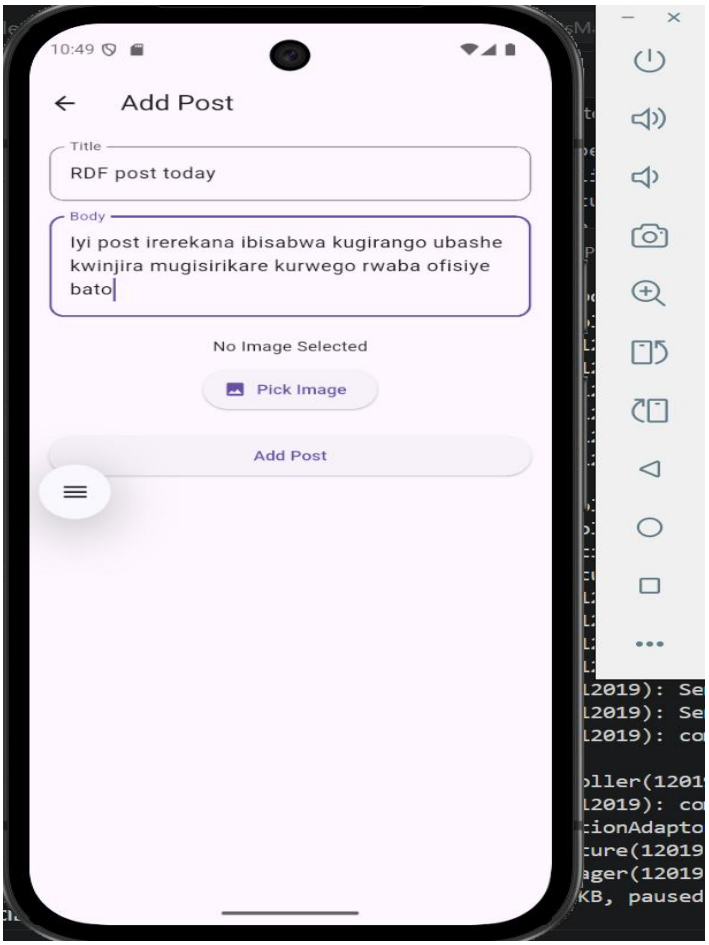
This screenshot Shows the full details of a selected post.

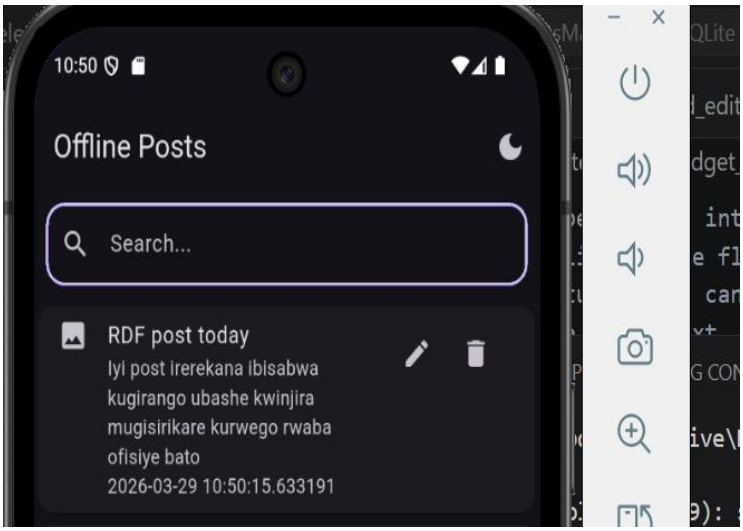



10:45
Details
Educational/ How to post(Authority)
there posts teach your audience something useful, establishing you as an expert.
2026-03-29 10:30:44.970810

3. Add Post

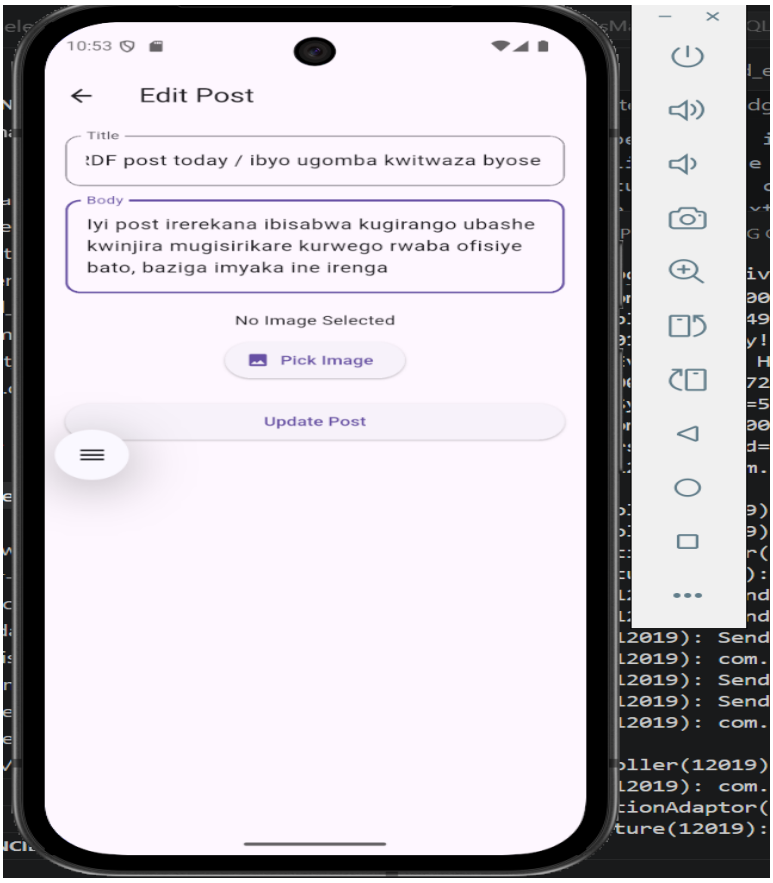
This screenshot shows that the user can create a new post.

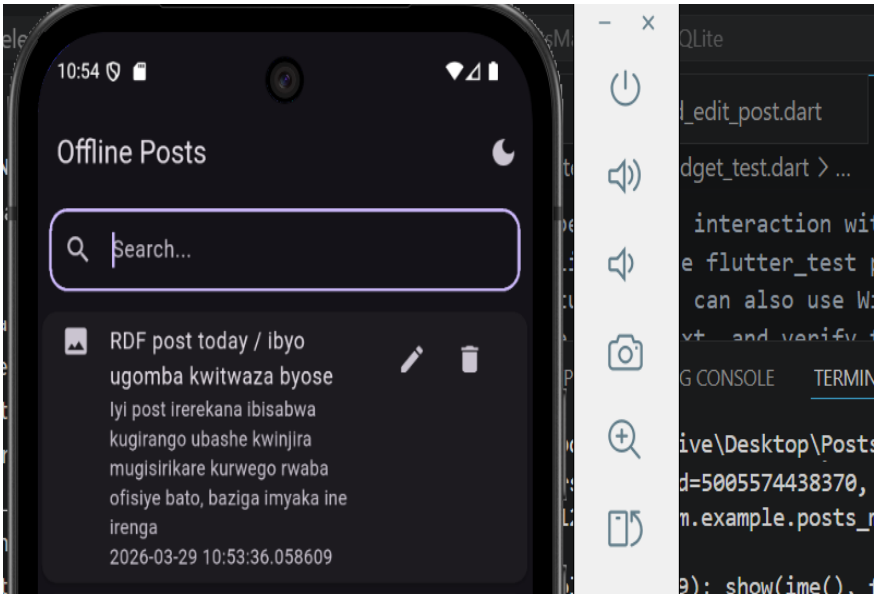




4. Edit Post

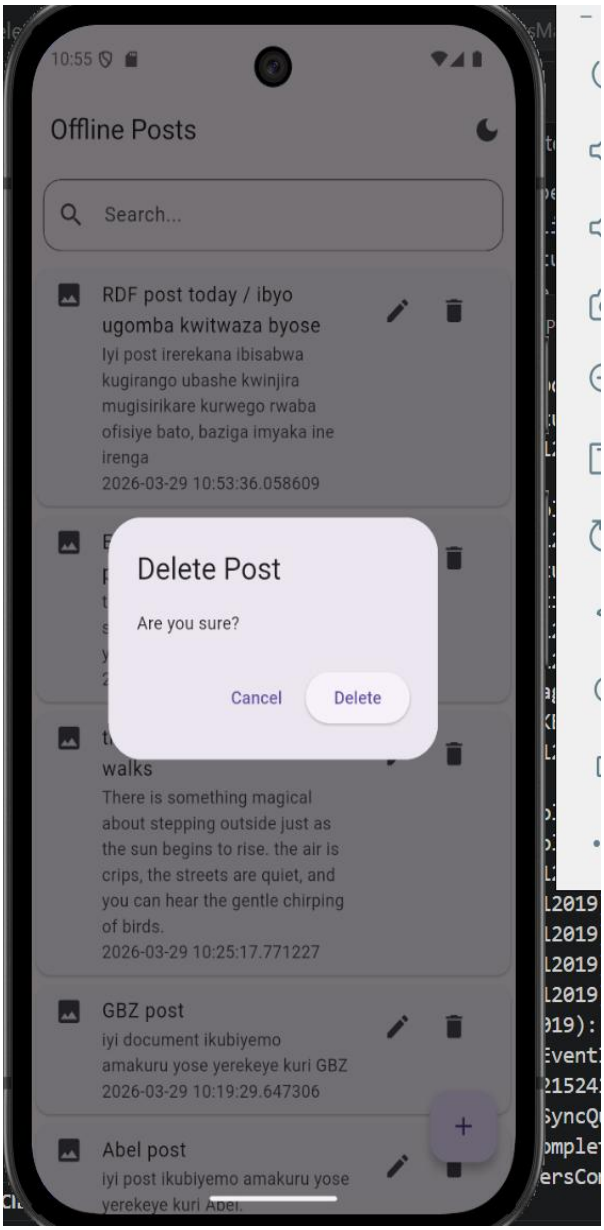
This screenshot shows that the user can update an existing post.

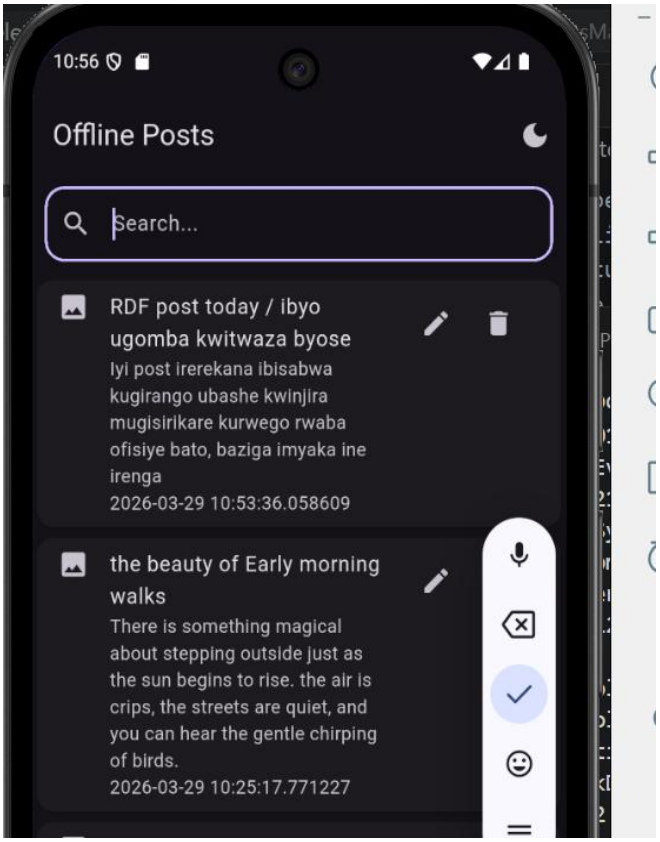


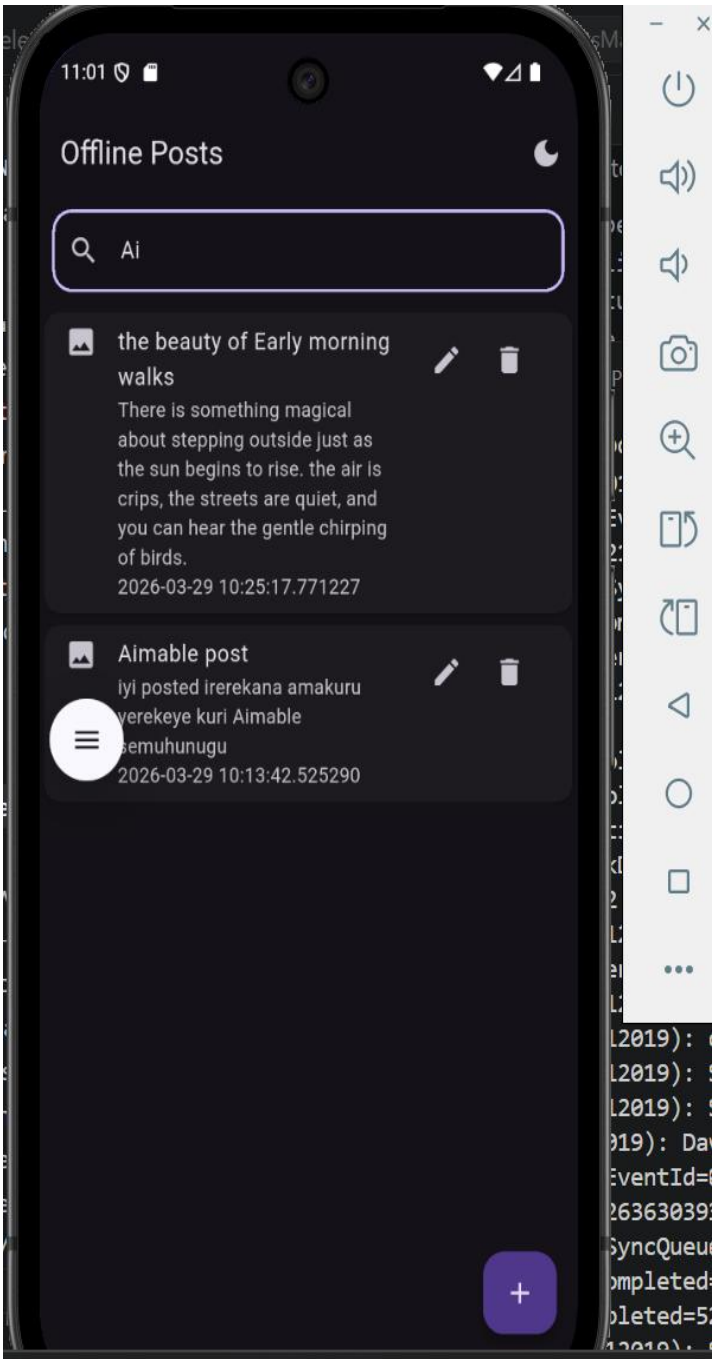


5. Delete Confirmation

This screenshot shows that the system asks the user to confirm before deleting a post.




10:56
Offline Posts
Search...
RDF post today / ibyo ugomba kwitwaza byose
Iyi post irerekana ibisabwa kugirango ubashe kwinjira mugisirikare kurwego rwaba ofisiye bato, baziga imyaka ine irenga
2026-03-29 10:53:36.058609
the beauty of Early morning walks
There is something magical about stepping outside just as the sun begins to rise. the air is crips, the streets are quiet, and you can hear the gentle chirping of birds.
2026-03-29 10:25:17.771227

6. Searching post

This screenshot shows that user can search any post he/she want.